Ismail Zain Syed Ahamed

Senior Data Scientist & Machine Learning Engineer

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Websites & Social Links: **LinkedIn** **GitHub** **Portfolio**

Summary

Dynamic and dedicated Senior Data Scientist & Machine Learning Engineer specializing in Generative AI, LLMs, and Autonomous AI Systems with over 4 plus years of extensive experience in the data-driven field and in developing data-intensive applications, resolving complex architectural and scalability challenges and prompting AI Agents.

Experience:

Data Science Consultant (Free Lancing) – Intelligent Health Tech (IHT) - USA, 04/2026 – Present
Medical Claims Verification and Submission Pilot Bot

- Developed an end-to-end automation system bridging NextGen EHR and Carelon payer portals, utilizing an Electron/React frontend and a FastAPI (Python) backend to orchestrate complex Playwright browser automation workflows.
- Leveraged AWS Bedrock (Claude models) for intelligent document processing to accurately extract unstructured patient data and procedure codes from PDF order sheets, and engineered dynamic prompt systems to autonomously answer complex clinical MCQs based on EHR encounter context.
- Designed a state-machine-driven execution engine featuring strategic "human-in-the-loop" checkpoints (for MFA, document review, and data verification), robust error recovery logic, and a local pre-submission safety store to guarantee data persistence and prevent data loss during portal timeouts.
- Deployed a scalable AWS environment (EC2, RDS PostgreSQL, S3, KMS) for securely handling sensitive medical documents and PHI, ensuring strict data encryption (at rest and in transit) alongside comprehensive audit logging.

Real-Time Automated Database Translation Pipeline & CDC Microservice

- Architected an event-driven Change Data Capture (CDC) microservice and bulk ETL pipeline in Python to automatically detect, translate, and synchronize text data between source and target PostgreSQL databases in real-time using LISTEN/NOTIFY triggers and UPSERT mechanisms.
- Engineered a highly concurrent integration with AWS Translate, leveraging Thread Pool Executor and a dynamic rate-limiting algorithm that queries the AWS Service Quotas API to maximize throughput at 70% capacity, effectively preventing API throttling and optimizing translation speed.
- Optimized performance and ensured data integrity by implementing MD5 hash-based state comparison for deduplication and a hybrid (in-memory and database) caching system, drastically reducing redundant API calls and processing overhead.
- Designed a resilient architecture with robust error handling, incorporating an idempotent change processing system, a Dead Letter Queue (DLQ) with automated exponential backoff retries, and an interactive CLI for human-in-the-loop review of flagged translation anomalies.

B-Informative IT Services Pvt Ltd. (BIITS), Bengaluru, Karnataka, India

Department Head – Data Science, 08/2025 – 03/2026
Assistant Manager – Data Analytics, 08/2024 – 07/2025

- As the head of the data science department, I used to work with cross-functional teams to use Python, SQL, and PySpark on Azure Databricks to create data-driven solutions.
- I used Azure Data Factory to automate end-to-end pipelines when designing, optimising, and building predictive models, including forecasting, ensemble methods, and LLMs.
- In addition to overseeing Azure storage and data quality, I strategically led a heterogeneous group of scientists, data engineers, and AI/ML experts to provide high-impact business logic.

Nu-Pie Management Consultancy Services, **Bengaluru, Karnataka, India**

Senior Associate Data Scientist, 10/2023 – 07/2024
Associate Data Scientist, 10/2022 – 09/2023
Junior Data Scientist, 06/2022 – 09/2022

- Developed a sales forecasting model using a hybrid time series approach for a drug store chain.
- Processed data stored in Azure Datalake, including managing white noise through Fast Fourier Transform.
- Utilized both traditional time series models and advanced deep learning models such as LSTM and RNN for sales prediction.
- Built data pipelines using Azure Datafactory and conducted modelling on the Databricks platform.

- Successfully predicted the pharmaceutical firm's sales for the next two months with accurate evaluation metrics (MAPE, MAE).

Daily Sales Apportionment & Reconciliation, Liquid IV -Unilever Health & Well Being,

- The automation initiative for Daily Sales Apportionment and Reconciliation, driven by Azure Stack technologies—Datalake, Datafactory, Databricks—paired with PowerBI for advanced business intelligence.
- Implemented streamlined data management by establishing a SharePoint drop for raw input files from various sources/channels and mounting Azure Datalake for storage.
- Further, optimizing the data processing workflows through Databricks notebooks and Datafactory pipeline orchestration with event-based triggers for each channel.
- Data processed from various sources is stored as Delta files within the Datalake, and using the processed data merged with master inventory dataset while maintaining uniformity, only the required fields are stored in a SQL Table (Azure SQL or Delta Tables).
- Integrated PowerBi with the SQL Table view, establishing a dynamic connection for dashboard visualization & successfully streamlined the daily sales process, reducing manual efforts and ensuring accurate, up-to-date business intelligence for strategic decision-making.

GRS Industries, Bengaluru, Karnataka, India
Quality Control Head and Management Representative (Internship), 05/2018 – 07/2018

- Led a team responsible for inspecting and implementing measures in the development, improvement, and review of new specifications and procedures for products or processes.
- Successfully carried out and completed a project at G.R.S Industries titled 'Empirical study on G.R.S Industries' and Management of ISO 9001.

Education:

Master's in Data Science and Machine Learning, 2020-2022
PES University, Bengaluru, Karnataka, India

Bachelor of Technology in Electrical and Electronics Engineering, 2016-2020
Presidency University, Bengaluru, Karnataka, India

Skills:

- **Python**
- **R**
- **Machine Learning**
- **Azure Stack (Databricks, Datafactory**
- **GCP**
- **MLflow**
- **Git**
- **SAP Datasphere**
- **Statistics / Statistical Modelling**
- **Scikit-learn**
- **Docker**
- **SQL**
- **PySpark**
- **LLM's / Gen AI**
- **AWS**
- **Snowflake**
- **Tableau**
- **NLP**
- **CI/CD**
- **A/B Testing**
- **TensorFlow / PyTorch**
- **Kubernetes**

Notable Achievements:

- Engineered a multilingual, domain-adapted LLM pipeline to semantically interpret diverse diagnostic reports using cross-lingual embeddings, integrating Gemini as a 'GenAI' conversational layer.
- Developed and implemented robust business logics on Azure Databricks using Python, SQL, and PySpark, optimizing and fine-tuning predictive models for sales logistics.
- Streamlined daily sales processes, reducing manual efforts and ensuring accurate, up-to-date business intelligence through automated pipelines and comprehensive dashboards.
- Built an AI Agent on Relevance AI to auto-generate structured company summaries for sales and go-to-market teams, integrating dynamic web scraping and fine- tuning domain-specific LLMs.

Licenses & certifications:

- Databricks AI Agent Fundamentals Accreditation
- Databricks Generative AI Fundamentals Accreditation
- Databricks Fundamentals Accreditation
- Digital Transformation with Google Cloud